



MedTech Prototyping Skills (BME290L)

Syllabus

Instructor: Mark Palmeri

Office: 258 Hudson Hall Annex

Office Hours: <http://meet.palmeri.io>

Email: mark.palmeri@duke.edu

Lecture Times/Days/Location: Tues/Thurs (12:00-1:15 PM)
TBD

Lab Time/Day/Location: Thurs (1:45-4:45 PM) POD

Lab Master: Matt Brown

Email: matt.brown@duke.edu

TA: TBD

Email: TBD

TA: TBD

Email: TBD



Course Objectives

This course focuses on developing medical device prototyping skills that will be used in future project and design courses, with a focus on preparing our students for positions in the medtech industry. Students will work individually to complete design tasks that will be tested to quantitative specifications. Students will gain hands-on experience with device fabrication, debugging, testing and failure analysis.

Upon completion of this course, students should be able to:

- Perform functional decomposition and develop user action flow chart
- Utilize ECAD (KiCad) for:
 - Electronic schematic capture
 - Printed circuit board (PCB) layout
 - SPICE circuit simulation
- Modular breadboarding of circuits to translation to testable PCBs
- Design a battery-powered device that:
 - Accepts analog and digital user input
 - Outputs analog and digital outputs
 - Performs logic using discrete electronics (e.g., logic gates, 555 timers)
 - Passes testable specifications to 95% CI
- Implement electronic logic on microcontroller (ESP32 or equivalent platform)
 - Modular / testable code development

- Version control systems (git)
- Choose appropriate microcontroller:
 - GPIO pins
 - Memory
 - Data communication protocols
- Utilize CAD (Onshape) for:
 - Device enclosure design with UI/UX considerations
 - Input / Output considerations
 - Size / weight constraints
 - Preparation of mechanical drawings
- 3D printing of designs
 - Assembly of discrete enclosure pieces
 - Use of mechanical fasteners
- Perform Fault Tree Analysis (FTA)
- Maintain a design history file
- Write technical analysis documents / reports

Pre-/Co-requisites

- ECE101L (Pre)
- ECE110L (Co)

Course Schedule

The following table is an overview of all activities this semester.



Any significant changes to this schedule, including assignments and due dates, will be reflected here, on Gradescope, and will be announced in the General channel on Teams.

Schedule

Aa Name	 Dates	 Type	 Status	 Location	 Link	 Attachments	 Notes
Syllabus Review; Functional Decomposition & User Action Flowcharts	@January 12, 2023	 Lecture	Not started	Lecture Room (TBD)			
Install [E]CAD / git tools on laptops	@January 12, 2023	Lab	Not started	POD			
Functional Decomposition & User Action Flowcharts	@January 12, 2023 → January 19, 2023	 Assignment	Not started	Gradescope			

Aa Name	📅 Dates	🔍 Type	⚙ Status	📍 Location	🔗 Link	📎 Attachments	📝 Notes
<u>CAD Mini-Tasks</u>	@January 17, 2023 → January 24, 2023	📌 Assignment	Not started	Gradescope			
<u>Intro to CAD Workflows (Flange)</u>	@January 17, 2023	📖 Lecture	Not started	Lecture Room (TBD)			
<u>CAD Drawings and Assemblies</u>	@January 19, 2023	📖 Lecture	Not started	Lecture Room (TBD)			
<u>CAD Tasks</u>	@January 19, 2023	🏢 Lab	Not started	POD			
<u>ECAD Schematic Capture</u>	@January 24, 2023 → January 31, 2023	📌 Assignment	Not started	Gradescope			
<u>Intro to ECAD (Schematic Capture)</u>	@January 24, 2023	📖 Lecture	Not started	Lecture Room (TBD)			
<u>ECAD Schematic Workflows</u>	@January 26, 2023	📖 Lecture	Not started	Lecture Room (TBD)			
<u>ECAD Schematics</u>	@January 26, 2023	🏢 Lab	Not started	POD			
<u>ECAD PCB Layout</u>	@January 31, 2023 → February 7, 2023	📌 Assignment	Not started	Gradescope			
<u>Intro to ECAD (PCB Layout)</u>	@January 31, 2023	📖 Lecture	Not started	Lecture Room (TBD)			
<u>PCB Design (Single- and Double-Layered Boards, SMT vs. TH)</u>	@February 2, 2023	📖 Lecture	Not started	Lecture Room (TBD)			
<u>PCB Layout</u>	@February 2, 2023	🏢 Lab	Not started	POD			
<u>IV Pump Func. Decomp. & User Action Flowchart</u>	@February 7, 2023 → February 14, 2023	📌 Assignment	Not started	Gradescope			
<u>IV Infusion Pump Project</u>	@February 7, 2023	📖 Lecture	Not started	Lecture Room (TBD)			
<u>IV Infusion Pump Project</u>	@February 9, 2023	📖 Lecture	Not started	Lecture Room (TBD)			
<u>IV Infusion Pump Func. Decomp. & User Action Flowchart</u>	@February 9, 2023	🏢 Lab	Not started	POD			
<u>IV Pump Schematic Capture</u>	@February 14, 2023 → February 21, 2023	📌 Assignment	Not started	Gradescope			
<u>IV Pump Discrete Electronic Design</u>	@February 14, 2023	📖 Lecture	Not started	Lecture Room (TBD)			

Aa Name	📅 Dates	🔍 Type	⚙ Status	📍 Location	🔗 Link	📎 Attachments	☰ Notes
IV Pump Schematic Capture	@February 16, 2023	Lecture	Not started	Lecture Room (TBD)			
IV Pump Schematic Capture	@February 16, 2023	Lab	Not started	POD			
Software Flowchart	@February 21, 2023 → February 28, 2023	Assignment	Not started	Gradescope			
Microcontrollers: How to Choose	@February 21, 2023	Lecture	Not started	Lecture Room (TBD)			
Firmware Development	@February 23, 2023	Lecture	Not started	Lecture Room (TBD)			
Discrete Component Breadboard Testing & Software Flowcharts	@February 23, 2023	Lab	Not started	POD			
Firmware Dev: Interrupts	@February 28, 2023	Lecture	Not started	Lecture Room (TBD)			
IV Pump PCB Design	@March 2, 2023 → March 9, 2023	Assignment	Not started	Gradescope			
Firmware Dev	@March 2, 2023	Lecture	Not started	Lecture Room (TBD)			
IV Pump PCB Design & Firmware Testing	@March 2, 2023	Lab	Not started	POD			
PCB Design	@March 7, 2023	Lecture	Not started	Lecture Room (TBD)			
PCB Design	@March 9, 2023	Lecture	Not started	Lecture Room (TBD)			
PCB Finalization (submit to mil)	@March 9, 2023	Lab	Not started	POD			
🌸 Spring Break	@March 14, 2023		Not started				NO CLASS
🌸 Spring Break	@March 16, 2023		Not started				NO CLASS
Testing to Spec Assignment (Python)	@March 21, 2023 → March 28, 2023	Assignment	Not started	Gradescope			
Testing to Specification (95% CI)	@March 21, 2023	Lecture	Not started	Lecture Room (TBD)			
IV Pump CAD Design	@March 23, 2023 → March 30, 2023	Assignment	Not started	Gradescope			
CAD Design	@March 23, 2023	Lecture	Not started	Lecture Room (TBD)			
PCB Testing	@March 23, 2023	Lab	Not started	POD			

Aa Name	📅 Dates	📌 Type	☀ Status	📍 Location	🔗 Link	📎 Attachments	☰ Notes
Fault Tree Analyses	@March 28, 2023 → April 4, 2023	📌 Assignment	Not started	Gradescope			
Fault Tree Analyses	@March 28, 2023	📖 Lecture	Not started	Lecture Room (TBD)			
Fault Tree Analyses	@March 30, 2023	📖 Lecture	Not started	Lecture Room (TBD)			
PCB Testing	@March 30, 2023	🔬 Lab	Not started	POD			
CAD Revisions (3D Printing)	@April 4, 2023	📖 Lecture	Not started	Lecture Room (TBD)			
Device Integration Redesign	@April 6, 2023	📖 Lecture	Not started	Lecture Room (TBD)			
Device Integration	@April 6, 2023	🔬 Lab	Not started	POD			
Integration Time	@April 11, 2023	📖 Lecture	Not started	Lecture Room (TBD)			
Integration Time	@April 13, 2023	📖 Lecture	Not started	Lecture Room (TBD)			
Integrated Device Testing	@April 13, 2023	🔬 Lab	Not started	POD			
Careers in MedTech	@April 18, 2023	📖 Lecture	Not started	Lecture Room (TBD)			
📄 Final Design Reports Due	@April 24, 2023 9:00 AM	📌 Assignment	Not started	Gradescope			

Textbooks & Resources

Texts & Resources

Aa Name	☰ Authors / Provider	🔗 Link	☰ Type	📌 Required / Optional	☰
KiCad		https://kicad.org/	ECAD	Required	Be dc la ve 6.
Onshape		https://duke.onshape.com	CAD	Required	Yc in Di er
Teams		https://teams.microsoft.com	Communication	Required	W by P; te
Practical Electronics for Inventors	Scherz & Monk	https://find.library.duke.edu/catalog/DUKE009381868	Text	Required	M be Di wl cē

Aa Name	≡ Authors / Provider	🔗 Link	≡ Type	⊕ Required / Optional	≡
Product Design and Development	Ulrich & Sppinger	http://www.ulrich-eppinger.net/	Text	Optional	
Design of Biomedical Devices and Systems	King, Fries & Johnson	https://www.crcpress.com/Design-of-Biomedical-Devices-and-Systems/King-Fries-Johnson/p/book/9781466569133	Text	Optional	
Engineering by Design	Voland	https://www.pearson.com/us/higher-education/program/Voland-Engineering-by-Design-2nd-Edition/PGM73262.html	Text	Optional	
Engineering Design: A project-based introduction	Dym, Little & Orwin	https://www.wiley.com/en-us/Engineering+Design%3A+A+Project+Based+Introduction%2C+4th+Edition-p-9781118324585	Text	Optional	
Handbook of Biological Statistics	McDonald	https://www.biostathandbook.com/	Text	Required	O ec st re
MeiserTask		https://meisertask.com	Kanban Board	Required	O Tr
Trello		https://trello.com	Kanban Board	Required	O M
diagrams.net (draw.io)		https://diagrams.net	Diagramming	Required	
Mindmeister		https://mindmeister.com	Brainstorming		
GitLab		https://gitlab.oit.duke.edu	VCS		Ri sc pr
What is a Kanban Board?		https://kanbanize.com/kanban-resources/getting-started/what-is-kanban-board	Kanban Board Reading		
Thingiverse		https://www.thingiverse.com	CAD		
NIH 3D Print Exchange		https://3dprint.nih.gov/	CAD		
Learn Scrum in Under 10 Min		https://www.axosoft.com/scrum	Agile Reading		
Functional Decomposition & Specifications	Ulrich & Eppinger		Reading	Required	

Attendance & Participation

Class participation, including lecture attendance and scheduled lab time, contributes to your class grade. Not being able to participate in class activities due to illness should be reported using the [Short Term Illness Form \(STIF\)](#) before the missed class activity.



If you feel any illness symptoms, do not show up to any in person activities and get in touch with Duke Student Health.



If you have a non-illness-related reason for not being able to participate in a class activity or meet an assignment submission deadline, please reach out to Dr. Palmeri as soon as possible to discuss the situation.



Grading

Breakdown

Participation: 15%
Midterm Assignments: 45%
Final Design Report: 40%

Final Course Grades

93% ≤ **A** ≤ 100%
90% ≤ **A-** < 93%
87% ≤ **B+** < 90%
83% ≤ **B** < 87%
...
<65% **F**

Gradescope

All graded assignments, associated due dates, and final grades will be posted on Gradescope and the [Syllabus Schedule](#). Gradescope will also be used to provide feedback and grading rubrics.



You must associate the pages of your submission with the grading rubric criteria during the submission process.
Failure to do so will result in lost assignment credit.

Late Assignment Submission Policy

Late work will be deducted 5% per twenty-four hour period that elapses after the due date. If foreseen or unforeseen circumstances prevent you from completing an assignment on time, you may request an extension. Extensions must be requested in advance of the due date. If the situation warrants an extension, we will determine a new due date for the essay based on your individual circumstances.

Unexcused late assignments will lose 50% of their potential point value for each 24 hour period beyond the due date (i.e., 100% → 50% → 25% . . .). Permission to submit an assignment late should be sought from Dr. Palmeri as far in advance as reasonably possible, but no less than 48 hours in advance, except in cases of illness.



All assignments must be satisfactorily completed by each student to pass the class, even if no credit will be awarded based on the late policy.

Regrades

Any regrading requests need to be made within one week of grades for a given assignment being returned using Gradescope. You must provide a description of why you feel a regrade is appropriate. Requesting a regrade could lead to additional loss of credit when a problem is re-evaluated.

Some assignments will have an opportunity to be resubmitted based on grading feedback at the discretion of Dr. Palmeri.



Duke Community Standard & Academic Honor

Engineering is inherently a collaborative field, and in this class, you are encouraged to work collaboratively on your projects. The work that you submit must be the product of your and your group's effort and understanding. All resources developed by another person or company, and used in your project, must be properly acknowledged.

All students are expected to adhere to all principles of the Duke Community Standard (<https://trinity.duke.edu/undergraduate/academic-policies/community-standard-student-conduct>). Violations of the Duke Community

Standard will be referred immediately to the Office of Student Conduct. Please do not hesitate to talk with Dr. Palmeri about any situations involving academic honor, especially if it is ambiguous what should be done.